

ReviewServlet.java

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package com.harisudhan.harisudhanmart.controller;

import com.harisudhan.harisudhanmart.exception.ValidationException;
import com.harisudhan.harisudhanmart.service.ReviewService;
import com.google.gson.JsonObject;
import java.io.IOException;
import java.sql.SQLException;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/** F8: product reviews and star ratings, restricted to buyers with a delivered order. */
@WebServlet(urlPatterns = {"/api/v1/reviews", "/api/v1/reviews/submit"})
public class ReviewServlet extends BaseServlet {

    @Override
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {
        ReviewService service = new ReviewService(daoFactory().reviewDAO());
        try {
            long productId = Long.parseLong(req.getParameter("productId"));
            writeJson(resp, HttpServletResponse.SC_OK, service.forProduct(productId));
        } catch (SQLException e) {
            writeError(resp, HttpServletResponse.SC_INTERNAL_SERVER_ERROR, "DB_ERROR", "Lookup failed");
        }
    }

    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse resp) throws ServletException, IOException {
        long userId = currentUserId(req.getSession(false));
        ReviewService service = new ReviewService(daoFactory().reviewDAO());
        StringBuilder sb = new StringBuilder();
        String line;
        while ((line = req.getReader().readLine()) != null) {
            sb.append(line);
        }
        JsonObject body = com.google.gson.JsonParser.parseString(sb.length() == 0 ? "{}" : sb.toString()).getAsJsonObject();

        try {
            long productId = body.get("productId").getAsLong();
            int rating = body.get("rating").getAsInt();
            String comment = body.has("comment") ? body.get("comment").AsString() : null;
            writeJson(resp, HttpServletResponse.SC_CREATED, service.submit(userId, productId, rating, comment));
        } catch (ValidationException e) {
            writeError(resp, HttpServletResponse.SC_BAD_REQUEST, e.getFieldErrorCode(), e.getMessage());
        } catch (SQLException e) {
            writeError(resp, HttpServletResponse.SC_INTERNAL_SERVER_ERROR, "DB_ERROR", "Review submission failed");
        }
    }
}
```